

Math 12 Curriculum

Intermediate Algebra | Middlesex School Mathematics

COURSE AT A GLANCE

Review Content

REVIEW

Primary resources: [1.1 Use the Language of Algebra](#); [1.3 Fractions](#); [1.5 Properties of Real Numbers](#)
Supplemental resources: [1.2 Integers](#)

UNIT LEARNING OBJECTIVES

- I can evaluate numerical expressions using the order of operations.
- I can perform operations with signed integers and fractions.
- I can simplify an algebraic expression by combining like terms.
- I can use the distributive property to rewrite an expression.
- I can determine the prime factorization of a whole number.

Linear Equations and Applications

REQUIRED

Primary resources: [2.1 Use a General Strategy to Solve Linear Equations](#); [2.3 Solve a Formula for a Specific Variable](#); [2.2 Use a Problem Solving Strategy](#); [2.4 Solve Mixture and Uniform Motion Applications](#)
Supplemental resources: [2.2 Use a Problem Solving Strategy](#)

UNIT LEARNING OBJECTIVES

- I can solve a linear equation and present equivalent steps clearly.
- I can solve a linear equation containing fractions.
- I can rearrange a formula to isolate a specified variable.
- I can translate a verbal relationship into a linear equation.
- I can solve and interpret a number, mixture, or motion problem using a linear equation.

Inequalities and Absolute-Value Equations

REQUIRED

Primary resources: [2.5 Solve Linear Inequalities](#); [2.6 Solve Compound Inequalities](#); [2.7 Solve Absolute Value Inequalities](#)

UNIT LEARNING OBJECTIVES

- I can solve and graph a one-variable linear inequality.
- I can solve a compound inequality involving AND.
- I can solve a compound inequality involving OR.
- I can express an inequality solution using interval notation.
- I can solve an absolute-value equation and check its solutions.

Lines and Introductory Functions

REQUIRED

Primary resources: [3.2 Slope of a Line](#); [3.1 Graph Linear Equations in Two Variables](#); [3.3 Find the Equation of a Line](#); [3.5 Relations and Functions](#)

UNIT LEARNING OBJECTIVES

- I can determine and interpret the slope of a line.
- I can graph a line from an equation in a common form.
- I can write an equation of a line from given information.
- I can determine whether two lines are parallel, perpendicular, or neither.
- I can evaluate a function using function notation.
- I can determine whether a relation is a function using the vertical line test or input-output data.
- I can determine domain and range from a graph or table.
- I can interpret domain and range in context.

Systems of Linear Equations

REQUIRED

Primary resources: [4.1 Solve Systems of Linear Equations with Two Variables](#); [4.2 Solve Applications with Systems of Equations](#)

UNIT LEARNING OBJECTIVES

- I can interpret the solution of a linear system as an intersection point.
- I can solve a two-variable linear system by graphing.
- I can solve a two-variable linear system by substitution.
- I can solve a two-variable linear system by elimination.
- I can create and solve a two-variable linear system for an application.

Polynomial Operations and Basic Factoring

REQUIRED

Primary resources: [5.2 Properties of Exponents and Scientific Notation](#); [5.1 Add and Subtract Polynomials](#); [5.3 Multiply Polynomials](#); [5.4 Dividing Polynomials](#); [6.1 Greatest Common Factor and Factor by Grouping](#)

UNIT LEARNING OBJECTIVES

- I can simplify expressions using integer exponent rules.
- I can add and subtract polynomials.
- I can multiply polynomials, including binomials.
- I can divide a polynomial by a monomial.
- I can factor a greatest common factor from a polynomial.

Extension Content

EXTENSION

Primary resources: [Intermediate Algebra 2e 2.7 Solve Absolute Value Inequalities](#); [Intermediate Algebra 2e 4.7 Graphing Systems of Linear Inequalities](#); [Intermediate Algebra 2e 6.2 Factor Trinomials](#); [Precalculus 2e 1.2 Domain and Range](#)

UNIT LEARNING OBJECTIVES

- I can solve and graph an absolute-value inequality.
- I can graph a system of linear inequalities and identify its solution region.
- I can factor a quadratic trinomial.
- I can identify a relationship as discrete or continuous.

IN-DEPTH CURRICULUM

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Review Content

REVIEW

OpenStax coverage: **Primary resources:** [1.1 Use the Language of Algebra](#); [1.3 Fractions](#); [1.5 Properties of Real Numbers](#)

Supplemental resources: [1.2 Integers](#)

M12-REV-001

I can evaluate numerical expressions using the order of operations.

OpenStax: **Primary resources:** [1.1 Use the Language of Algebra](#)

SUPPORTING SKILLS



Apply the order of operations to a numerical expression.

SK-NUM-ORDER-OPERATIONS · inherited from middle school

M12-REV-002

I can perform operations with signed integers and fractions.

OpenStax: **Primary resources:** [1.3 Fractions](#)

Supplemental resources: [1.2 Integers](#)

SUPPORTING SKILLS



Add, subtract, multiply, and divide fractions.

SK-NUM-FRACTION-OPERATE · inherited from middle school



Add, subtract, multiply, and divide signed integers.

SK-NUM-INTEGER-OPERATE · inherited from middle school

M12-REV-003

I can simplify an algebraic expression by combining like terms.

OpenStax: **Primary resources:** [1.1 Use the Language of Algebra](#)

SUPPORTING SKILLS

- R** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- R** Identify and combine like terms.
SK-ALG-LIKE-TERMS · inherited from middle school

M12-REV-004

I can use the distributive property to rewrite an expression.

OpenStax: **Primary resources:** [1.5 Properties of Real Numbers](#)

SUPPORTING SKILLS

- R** Apply the distributive property in either direction.
SK-ALG-DISTRIBUTE · inherited from middle school

M12-REV-005

I can determine the prime factorization of a whole number.

OpenStax: **Primary resources:** [1.1 Use the Language of Algebra](#)

SUPPORTING SKILLS

- R** Express a whole number as a product of prime factors.
SK-NUM-PRIME-FACTOR · inherited from middle school

Linear Equations and Applications

REQUIRED

OpenStax coverage: **Primary resources:** [2.1 Use a General Strategy to Solve Linear Equations](#); [2.3 Solve a Formula for a Specific Variable](#); [2.2 Use a Problem Solving Strategy](#); [2.4 Solve Mixture and Uniform Motion Applications](#)
Supplemental resources: [2.2 Use a Problem Solving Strategy](#)

M12-EQN-001

I can solve a linear equation and present equivalent steps clearly.

OpenStax: **Primary resources:** [2.1 Use a General Strategy to Solve Linear Equations](#)

SUPPORTING SKILLS

- I** Check a proposed solution in the original equation.
SK-ALG-CHECK-SOLUTION
- A** Apply the distributive property in either direction.
SK-ALG-DISTRIBUTE · inherited from middle school
- I** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS
- A** Identify and combine like terms.
SK-ALG-LIKE-TERMS · inherited from middle school

M12-EQN-002

I can solve a linear equation containing fractions.

OpenStax: **Primary resources:** [2.1 Use a General Strategy to Solve Linear Equations](#)

SUPPORTING SKILLS

- A** Check a proposed solution in the original equation.
SK-ALG-CHECK-SOLUTION · first introduced in Math 12
- I** Clear fractions from an equation using a common denominator.
SK-ALG-CLEAR-FRACTIONS
- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A** Add, subtract, multiply, and divide fractions.
SK-NUM-FRACTION-OPERATE · inherited from middle school

M12-EQN-003

I can rearrange a formula to isolate a specified variable.

OpenStax: **Primary resources:** [2.3 Solve a Formula for a Specific Variable](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Isolate a specified variable in an equation or formula.
SK-ALG-VARIABLE-ISOLATE

M12-EQN-004

I can translate a verbal relationship into a linear equation.

OpenStax: **Primary resources:** [2.2 Use a Problem Solving Strategy](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I** Translate quantities and relationships into an equation.
SK-ALG-TRANSLATE-EQUATION

I can solve and interpret a number, mixture, or motion problem using a linear equation.

OpenStax: Primary resources: [2.4 Solve Mixture and Uniform Motion Applications](#)

Supplemental resources: [2.2 Use a Problem Solving Strategy](#)

SUPPORTING SKILLS

- A** Check a proposed solution in the original equation.
SK-ALG-CHECK-SOLUTION · first introduced in Math 12
- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- D** Translate quantities and relationships into an equation.
SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12

Inequalities and Absolute-Value Equations

REQUIRED

OpenStax coverage: **Primary resources:** [2.5 Solve Linear Inequalities](#); [2.6 Solve Compound Inequalities](#); [2.7 Solve Absolute Value Inequalities](#)

M12-INE-001

I can solve and graph a one-variable linear inequality.

OpenStax: **Primary resources:** [2.5 Solve Linear Inequalities](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Represent an inequality solution on a number line.
SK-INE-NUMBER-LINE
- I** Reverse an inequality when multiplying or dividing by a negative value.
SK-INE-REVERSE-SIGN

M12-INE-002

I can solve a compound inequality involving AND.

OpenStax: **Primary resources:** [2.6 Solve Compound Inequalities](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Interpret AND as the intersection of two inequality solution sets.
SK-INE-COMPOUND-AND
- A** Represent an inequality solution on a number line.
SK-INE-NUMBER-LINE · first introduced in Math 12
- A** Reverse an inequality when multiplying or dividing by a negative value.
SK-INE-REVERSE-SIGN · first introduced in Math 12

M12-INE-003

I can solve a compound inequality involving OR.

OpenStax: **Primary resources:** [2.6 Solve Compound Inequalities](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Interpret OR as the union of two inequality solution sets.
SK-INE-COMPOUND-OR
- A** Represent an inequality solution on a number line.
SK-INE-NUMBER-LINE · first introduced in Math 12
- A** Reverse an inequality when multiplying or dividing by a negative value.
SK-INE-REVERSE-SIGN · first introduced in Math 12

M12-INE-004

I can express an inequality solution using interval notation.

OpenStax: **Primary resources:** [2.6 Solve Compound Inequalities](#)

SUPPORTING SKILLS

- A** Represent an inequality solution on a number line.
SK-INE-NUMBER-LINE · first introduced in Math 12
- I** Read inequality symbols and state the relationship each symbol expresses.
SK-NOT-INEQUALITY-READ
- I** Translate between inequality and interval notation.
SK-NOT-INTERVAL

M12-INE-005

I can solve an absolute-value equation and check its solutions.

OpenStax: **Primary resources:** [2.7 Solve Absolute Value Inequalities](#)

SUPPORTING SKILLS

- I** Rewrite an absolute-value equation as two cases.
SK-ABS-SPLIT-EQUATION
- A** Check a proposed solution in the original equation.
SK-ALG-CHECK-SOLUTION · first introduced in Math 12
- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12

Lines and Introductory Functions

REQUIRED

OpenStax coverage: **Primary resources:** [3.2 Slope of a Line](#); [3.1 Graph Linear Equations in Two Variables](#); [3.3 Find the Equation of a Line](#); [3.5 Relations and Functions](#)

M12-LIN-001

I can determine and interpret the slope of a line.

OpenStax: **Primary resources:** [3.2 Slope of a Line](#)

SUPPORTING SKILLS

- I** Determine slope from a graph.
SK-LIN-SLOPE-GRAPH
- I** Calculate slope from two points.
SK-LIN-SLOPE-POINTS
- I** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS

M12-LIN-002

I can graph a line from an equation in a common form.

OpenStax: **Primary resources:** [3.1 Graph Linear Equations in Two Variables](#)

SUPPORTING SKILLS

- I** Convert among common forms of a linear equation.
SK-LIN-FORM-CONVERT
- I** Use slope to generate additional points on a line.
SK-LIN-GRAPH-SLOPE
- I** Plot an intercept accurately on a coordinate plane.
SK-LIN-PLOT-INTERCEPT

M12-LIN-003

I can write an equation of a line from given information.

OpenStax: **Primary resources:** [3.3 Find the Equation of a Line](#)

SUPPORTING SKILLS

- A** Isolate a specified variable in an equation or formula.
SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- D** Convert among common forms of a linear equation.
SK-LIN-FORM-CONVERT · first introduced in Math 12
- A** Calculate slope from two points.
SK-LIN-SLOPE-POINTS · first introduced in Math 12

M12-LIN-004

I can determine whether two lines are parallel, perpendicular, or neither.

OpenStax: **Primary resources:** [3.3 Find the Equation of a Line](#)

SUPPORTING SKILLS

- I** Recognize that distinct parallel lines have equal slopes.
SK-LIN-PARALLEL-SLOPE
- I** Determine the negative reciprocal of a nonzero slope.
SK-LIN-PERPENDICULAR-SLOPE

M12-LIN-005

I can evaluate a function using function notation.

OpenStax: **Primary resources:** [3.5 Relations and Functions](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT
- I** Identify the input and output represented by function notation.
SK-FUNC-NOTATION-READ
- I** Recall and correctly use the terms relation, function, input, output, domain, and range.
SK-FUNC-VOCABULARY

M12-LIN-006

I can determine whether a relation is a function using the vertical line test or input-output data.

OpenStax: **Primary resources:** [3.5 Relations and Functions](#)

SUPPORTING SKILLS

- D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- I** Apply the vertical line test to a graph.
SK-FUNC-VERTICAL-LINE
- A** Recall and correctly use the terms relation, function, input, output, domain, and range.
SK-FUNC-VOCABULARY · first introduced in Math 12

M12-LIN-008

I can determine domain and range from a graph or table.

OpenStax: **Primary resources:** [3.5 Relations and Functions](#)

SUPPORTING SKILLS

- I** Read domain and range from a graph.
SK-FUNC-DOMAIN-RANGE-GRAPH
- I** Read domain and range from a table.
SK-FUNC-DOMAIN-RANGE-TABLE

M12-LIN-009

I can interpret domain and range in context.

OpenStax: **Primary resources:** [3.5 Relations and Functions](#)

SUPPORTING SKILLS

- I** Restrict domain to values meaningful in a context.
SK-FUNC-CONTEXT-DOMAIN
- I** Interpret the range of a function in a context.
SK-FUNC-CONTEXT-RANGE
- I** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN

Systems of Linear Equations

REQUIRED

OpenStax coverage: **Primary resources:** [4.1 Solve Systems of Linear Equations with Two Variables](#); [4.2 Solve Applications with Systems of Equations](#)

M12-SYS-001

I can interpret the solution of a linear system as an intersection point.

OpenStax: **Primary resources:** [4.1 Solve Systems of Linear Equations with Two Variables](#)

SUPPORTING SKILLS

- I** Locate and interpret the intersection of two lines.

SK-SYS-GRAPH-INTERSECTION

M12-SYS-002

I can solve a two-variable linear system by graphing.

OpenStax: **Primary resources:** [4.1 Solve Systems of Linear Equations with Two Variables](#)

SUPPORTING SKILLS

- A** Use slope to generate additional points on a line.

SK-LIN-GRAPH-SLOPE · first introduced in Math 12

- A** Plot an intercept accurately on a coordinate plane.

SK-LIN-PLOT-INTERCEPT · first introduced in Math 12

- I** Check an ordered pair in both equations of a system.

SK-SYS-CHECK

- D** Locate and interpret the intersection of two lines.

SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12

M12-SYS-003

I can solve a two-variable linear system by substitution.

OpenStax: **Primary resources:** [4.1 Solve Systems of Linear Equations with Two Variables](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.

SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12

- A** Check an ordered pair in both equations of a system.

SK-SYS-CHECK · first introduced in Math 12

- I** Substitute an equivalent expression for a variable.

SK-SYS-SUBSTITUTE

M12-SYS-004

I can solve a two-variable linear system by elimination.

OpenStax: **Primary resources:** [4.1 Solve Systems of Linear Equations with Two Variables](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.

SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12

- A** Check an ordered pair in both equations of a system.

SK-SYS-CHECK · first introduced in Math 12

- I** Add equations to eliminate a variable.

SK-SYS-ELIMINATE

I can create and solve a two-variable linear system for an application.

OpenStax: **Primary resources:** [4.2 Solve Applications with Systems of Equations](#)

SUPPORTING SKILLS

- A** Check an ordered pair in both equations of a system.
SK-SYS-CHECK · first introduced in Math 12
- A** Add equations to eliminate a variable.
SK-SYS-ELIMINATE · first introduced in Math 12
- I** Define two variables and write two equations for a context.
SK-SYS-MODEL
- A** Substitute an equivalent expression for a variable.
SK-SYS-SUBSTITUTE · first introduced in Math 12

Polynomial Operations and Basic Factoring

REQUIRED

OpenStax coverage: **Primary resources:** [5.2 Properties of Exponents and Scientific Notation](#); [5.1 Add and Subtract Polynomials](#); [5.3 Multiply Polynomials](#); [5.4 Dividing Polynomials](#); [6.1 Greatest Common Factor and Factor by Grouping](#)

M12-POL-001

I can simplify expressions using integer exponent rules.

OpenStax: **Primary resources:** [5.2 Properties of Exponents and Scientific Notation](#)

SUPPORTING SKILLS

- I** Apply the power rule for exponents.
SK-EXP-POWER
- I** Apply the product rule for exponents.
SK-EXP-PRODUCT
- I** Apply the quotient rule for exponents.
SK-EXP-QUOTIENT

M12-POL-002

I can add and subtract polynomials.

OpenStax: **Primary resources:** [5.1 Add and Subtract Polynomials](#)

SUPPORTING SKILLS

- A** Identify and combine like terms.
SK-ALG-LIKE-TERMS · inherited from middle school
- I** Identify like polynomial terms by variable part and exponent.
SK-POLY-LIKE-TERMS
- I** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY

M12-POL-003

I can multiply polynomials, including binomials.

OpenStax: **Primary resources:** [5.3 Multiply Polynomials](#)

SUPPORTING SKILLS

- A** Apply the distributive property in either direction.
SK-ALG-DISTRIBUTE · inherited from middle school
- I** Multiply two binomials using distribution.
SK-POLY-BINOMIAL-MULTIPLY
- I** Multiply a monomial by a polynomial.
SK-POLY-DISTRIBUTE
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M12-POL-004

I can divide a polynomial by a monomial.

OpenStax: **Primary resources:** [5.4 Dividing Polynomials](#)

SUPPORTING SKILLS

- A** Apply the quotient rule for exponents.
SK-EXP-QUOTIENT · first introduced in Math 12
- I** Divide each term of a polynomial by a monomial.
SK-POLY-DIVIDE-MONOMIAL
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M12-POL-005

I can factor a greatest common factor from a polynomial.

OpenStax: **Primary resources:** [6.1 Greatest Common Factor and Factor by Grouping](#)

SUPPORTING SKILLS

- A** Apply the distributive property in either direction.
SK-ALG-DISTRIBUTE · inherited from middle school
- I** Factor the greatest common monomial factor from a polynomial.
SK-FACTOR-GCF
- I** Determine the numerical and variable greatest common factor of polynomial terms.
SK-POLY-GCF
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

Extension Content

EXTENSION

OpenStax coverage: **Primary resources:** [Intermediate Algebra 2e 2.7 Solve Absolute Value Inequalities](#); [Intermediate Algebra 2e 4.7 Graphing Systems of Linear Inequalities](#); [Intermediate Algebra 2e 6.2 Factor Trinomials](#); [Precalculus 2e 1.2 Domain and Range](#)

M12-EXT-001

I can solve and graph an absolute-value inequality.

OpenStax: **Primary resources:** [2.7 Solve Absolute Value Inequalities](#)

SUPPORTING SKILLS

- I** Rewrite an absolute-value inequality as a compound inequality.
SK-ABS-SPLIT-INEQUALITY
- A** Interpret AND as the intersection of two inequality solution sets.
SK-INE-COMPOUND-AND · first introduced in Math 12
- A** Interpret OR as the union of two inequality solution sets.
SK-INE-COMPOUND-OR · first introduced in Math 12
- A** Represent an inequality solution on a number line.
SK-INE-NUMBER-LINE · first introduced in Math 12

M12-EXT-002

I can graph a system of linear inequalities and identify its solution region.

OpenStax: **Primary resources:** [4.7 Graphing Systems of Linear Inequalities](#)

SUPPORTING SKILLS

- I** Graph the boundary line of a linear inequality with the correct line type.
SK-INE-GRAPH-BOUNDARY
- I** Identify the overlapping solution region of multiple linear inequalities.
SK-INE-SYSTEM-REGION
- A** Use slope to generate additional points on a line.
SK-LIN-GRAPH-SLOPE · first introduced in Math 12

M12-EXT-003

I can factor a quadratic trinomial.

OpenStax: **Primary resources:** [6.2 Factor Trinomials](#)

SUPPORTING SKILLS

- I** Verify a factorization by multiplication.
SK-FACTOR-CHECK
- I** Factor a quadratic trinomial with a nonunit leading coefficient.
SK-FACTOR-TRINOMIAL-GENERAL
- I** Factor a monic quadratic trinomial.
SK-FACTOR-TRINOMIAL-MONIC

M12-EXT-004

I can identify a relationship as discrete or continuous.

OpenStax: **Primary resources:** [1.2 Domain and Range](#)

SUPPORTING SKILLS

- I** Distinguish discrete inputs from values varying continuously over an interval.
SK-FUNC-DISCRETE-CONTINUOUS